

```
In [1]: import numpy as np
```

```
In [2]: arr = np.array([2, 4, 6, 8, 10]) # 1D array
arr2 = np.array([[2, 4, 6], [8, 10, 12]]) # 2D array
arr3 = np.array([[[1, 2], [3, 4]], [[5, 6], [7, 8]]]) # 3D array

# Elements in the array
print(arr.size)
print(arr2.size)
print(arr3.size)

print("\n")
# Dimensions of the array
print(arr.ndim)
print(arr2.ndim)
print(arr3.ndim)
```

5
6
8

1
2
3

```
In [3]: print("Shape", arr.shape)

print("Reshaping the array")
arr.reshape(1, 5)
```

Shape (5,)
Reshaping the array

```
Out[3]: array([[ 2,  4,  6,  8, 10]])
```

```
In [4]: arr_copy = arr.copy()
print("Original array:", arr)
print("Copied array:", arr_copy)
```

Original array: [2 4 6 8 10]
Copied array: [2 4 6 8 10]

```
In [5]: demo = arr2.ravel()
print("Ravelled array:", demo)
demo[1] = 20
print("Modified ravelled array:", demo)
```

Ravelled array: [2 4 6 8 10 12]
Modified ravelled array: [2 20 6 8 10 12]

```
In [6]: print(arr.sum()) # Sum of all elements in the array
print(arr2.sum(axis=1)) # Sum of each row in the array
```

30
[28 30]

```
In [7]: # Broadcasting
print(arr + 2) # Add 2 to each element
```

```
[ 4  6  8 10 12]
```

```
In [8]: # Masking
print(arr[arr > 5]) # Elements greater than 5
```

```
[ 6  8 10]
```

```
In [9]: # Where
print(np.where(arr > 5)) # positions of elements greater than 5
(array([2, 3, 4]),)
```

```
In [10]: marks = np.array([72, 48, 31, 81, 96, 55, 88, 42, 91, 67, 35, 74, 63, 39,
                             85, 78, 93, 69, 52])
grades = np.select(
    [marks >= 90, marks >= 70, marks >= 50, marks >= 33],
    ["A", "B", "C", "D"],
    default="F"
)

print("Grades for the marks are:")
for mark, grade in zip(marks, grades):
    print(f"Mark: {mark} - Grade: {grade}")
```

Grades for the marks are:

```
Mark: 72 - Grade: B
Mark: 48 - Grade: D
Mark: 31 - Grade: F
Mark: 81 - Grade: B
Mark: 96 - Grade: A
Mark: 55 - Grade: C
Mark: 88 - Grade: B
Mark: 42 - Grade: D
Mark: 91 - Grade: A
Mark: 67 - Grade: C
Mark: 35 - Grade: D
Mark: 74 - Grade: B
Mark: 63 - Grade: C
Mark: 39 - Grade: D
Mark: 85 - Grade: B
Mark: 78 - Grade: B
Mark: 93 - Grade: A
Mark: 69 - Grade: C
Mark: 52 - Grade: C
```

```
In [11]: import pandas as pd
```

```
In [12]: data = {
    'Name': ['Gourav', 'Sahil', 'Rudra', 'Kunal', 'Neeraj'],
    'Age': [21, 21, 19, 20, 18],
    'City': ['Delhi', 'Jaipur', 'Gurugram', 'Bhopal', 'Pune']
}

df = pd.DataFrame(data)
print(df)
```

	Name	Age	City
0	Gourav	21	Delhi
1	Sahil	21	Jaipur
2	Rudra	19	Gurugram
3	Kunal	20	Bhopal
4	Neeraj	18	Pune

```
In [13]: print("first two rows of the dataframe are: ")
print(df.head(2))
print('\n')
print("last two rows of the dataframe are: ")
print(df.tail(2))
```

first two rows of the dataframe are:

	Name	Age	City
0	Gourav	21	Delhi
1	Sahil	21	Jaipur

last two rows of the dataframe are:

	Name	Age	City
3	Kunal	20	Bhopal
4	Neeraj	18	Pune

```
In [14]: print("eliminating a column from the dataframe: ")
print(df.drop("City", axis=1))
print("\nEliminating a row from the dataframe: ")
print(df.drop(index=4, axis=0))
```

eliminating a column from the dataframe:

	Name	Age
0	Gourav	21
1	Sahil	21
2	Rudra	19
3	Kunal	20
4	Neeraj	18

Eliminating a row from the dataframe:

	Name	Age	City
0	Gourav	21	Delhi
1	Sahil	21	Jaipur
2	Rudra	19	Gurugram
3	Kunal	20	Bhopal

```
In [15]: print(df.loc[0])
```

```
Name    Gourav
Age      21
City    Delhi
Name: 0, dtype: object
```

```
In [16]: marks = pd.DataFrame({
    'Marks' : [72, 88, 65, 91, 54]
})

df = pd.merge(df, marks, left_index=True, right_index=True)
print(df)
```

	Name	Age	City	Marks
0	Gourav	21	Delhi	72
1	Sahil	21	Jaipur	88
2	Rudra	19	Gurugram	65
3	Kunal	20	Bhopal	91
4	Neeraj	18	Pune	54

```
In [17]: print("Description of the DataFrame:", df.describe())
```

```
Description of the DataFrame:           Age      Marks
count    5.000000    5.000000
mean     19.800000   74.000000
std       1.30384    15.572412
min      18.000000   54.000000
25%      19.000000   65.000000
50%      20.000000   72.000000
75%      21.000000   88.000000
max      21.000000   91.000000
```

```
In [18]: print("Check for null values: ", df.isnull().sum())
```

```
Check for null values:  Name      0
Age      0
City      0
Marks     0
dtype: int64
```

```
In [19]: df['Grade'] = df['Marks'].apply(lambda x: 'A' if x >= 90 else ('B' if x >
print(df))
```

	Name	Age	City	Marks	Grade
0	Gourav	21	Delhi	72	C
1	Sahil	21	Jaipur	88	B
2	Rudra	19	Gurugram	65	C
3	Kunal	20	Bhopal	91	A
4	Neeraj	18	Pune	54	D

```
In [20]: df1 = pd.DataFrame({'a': [None, 4, 4], 'b' : [2, 3, None]})
# Fill null values with 0

df1.fillna(0, inplace=True)
print(df1)
```

	a	b
0	0.0	2.0
1	4.0	3.0
2	4.0	0.0

```
In [21]: df2 = pd.DataFrame({
    'Name': ['Gourav', 'Sahil', 'Rudra', 'Kunal', 'Kunal']
})

print("Drop duplicates from the DataFrame:")
print(df2.drop_duplicates())
```

```
Drop duplicates from the DataFrame:
Name
0  Gourav
1  Sahil
2  Rudra
3  Kunal
```

```
In [22]: print("Rename columns:\n", df['Name'].rename('Full Name'))
```

Rename columns:

```
0    Gourav
1    Sahil
2    Rudra
3    Kunal
4    Neeraj
```

Name: Full Name, dtype: object

```
In [23]: print(df['Marks'].mean())
print(df['Age'].mean())
```

74.0

19.8

```
In [24]: df2 = pd.DataFrame({
    'Name': ['Gourav', 'Sahil', 'Rudra', 'Kunal', 'Neeraj'],
    'Groups': ['A', 'B', 'A', 'B', 'C'],
})

print(df2.groupby('Groups')['Name'].count())
```

Groups

A 2

B 2

C 1

Name: Name, dtype: int64

```
In [25]: # Saving the DataFrame into a CSV file
df.to_csv('output.csv', index=False)

# Loading the CSV file into a DataFrame
df_loaded = pd.read_csv('output.csv')
print("Loaded DataFrame from CSV:\n", df_loaded)
```

Loaded DataFrame from CSV:

	Name	Age	City	Marks	Grade
0	Gourav	21	Delhi	72	C
1	Sahil	21	Jaipur	88	B
2	Rudra	19	Gurugram	65	C
3	Kunal	20	Bhopal	91	A
4	Neeraj	18	Pune	54	D